

Technical Debt Analysis & Prioritized Cleanup Plan

Project: st185-link-ecu-config — Link G4X XtremeX ECU / RealDash / CAN bus configuration for 1993 Toyota Celica GT-Four ST185 TrackCluster

Repo: github.com/living-relation/st185-link-ecu-config

Analysis date: 2026-09-27

Local HEAD: b265f1d (up to date with origin/main)

Executive Summary

This project is a well-structured automotive CAN bus configuration repository with bench tooling, harness validation, and RealDash dashboard specs. The code quality is generally good — Python files are clean, well-documented, and follow consistent conventions. However, there are meaningful gaps in **test coverage, CI/CD, dependency management, and stale file cleanup** that should be addressed to prevent future drift and reduce onboarding friction.

Debt scorecard:

Category	Severity	Items
Stale/duplicate files	● High	10 versioned .rd files, duplicate CLAUDE.md/AGENTS.md content, stale .pyc artifacts
Missing CI/CD	● High	No linting, no tests, no dependency scanning, no harness validation in CI
No test coverage	● Medium	Zero unit tests across all Python code (bench, sender app, harness checks)
Documentation drift	● Medium	Handoff note committed (violates own rule), stale references to removed apps
Dependency management	● Low	No lock file, no Dependabot, Python version mismatch (3.12 CI vs 3.14 local)
Code quality	● Low	No type checking, no formatter, minor typing inconsistencies
Security	● Minimal	FAILSAFE=False on PyAutoGUI, no secrets exposed

1. Stale & Duplicate Files (Priority: ● High)

1.1 Versioned .rd files in rd-build/realdash-root/

Location: rd-build/realdash-root/

Files: st185_dash.rd , st185_dash_v2.rd through st185_dash_v10.rd (10 files total)

These are old RealDash dashboard snapshots from iterative builds. The current authoritative file is st185_dash.rd at the repo root. The 9 versioned copies in rd-build/realdash-root/ serve no purpose — git history already preserves every version.

Action: Delete all 9 `st185_dash_v*.rd` files from `rd-build/realdash-root/`. Keep only `st185_dash.rd` (the current root copy). If the `rd-build/realdash-root/` directory then contains no needed files, remove it entirely.

1.2 Committed pycache directories

Locations:

- `bench/__pycache__/`
- `apps/trackcluster-can-sender/__pycache__/` (contains `app.cpython-314.pyc`)
- `docs/harness/__pycache__/`

`.gitignore` excludes `__pycache__/` but these were committed before the rule was added. They remain tracked in git history and clutter the working tree.

Action: Remove all `__pycache__` directories from git tracking: `git rm -r --cached __pycache__/` in each location. The `.gitignore` rule will prevent re-addition.

1.3 Duplicate CLAUDE.md / AGENTS.md content

Files: `CLAUDE.md` (143 lines), `AGENTS.md` (149 lines)

These files share ~60% identical content: the "Related Repos" section, "Fast Commands", "Generated files" table, "Handoffs are never committed" policy, and "Model" pinning section are duplicated verbatim.

Action: Extract shared content into a single source (e.g., a `CONTRIBUTING.md` or `.claude/shared-rules.md`) and have both files reference it. Alternatively, make `AGENTS.md` the authoritative agent-facing file and slim `CLAUDE.md` to only Claude-specific overrides. This reduces the maintenance surface — currently, an update must be made in both files or they drift.

1.4 rd-build/tools/ one-off scripts

Location: `rd-build/tools/`

Files: `add_gauge.py`, `bind_pill.py`, `combo.py`, `fix_dyncolor.py`, `label_paste.py`, `paste_batch.py`, `rebind.py`, `set_levels.py`, `set_text.py`, `set_xywh.py`, `value_paste_bind.py`

These 11 scripts appear to be one-off RealDash editor automation helpers from the initial dashboard build process. If the dashboard build is complete, they are stale tooling that will confuse future contributors.

Action: Review each script. If no longer needed, move to `archive/2026-09-25-cleanup/` or delete. If some are still useful, document which ones in `rd-build/tools/SETUP.md`.

2. Missing CI/CD (Priority: ● High)

2.1 No code quality CI

The only GitHub Action (`research-hub.yml`) regenerates documentation HTML. There is no CI for:

- **Python linting** (ruff or flake8)
- **Type checking** (mypy or pyright)
- **Compilation check** (`python -m py_compile`)
- **Unit tests** (pytest)
- **Harness validation** (`python docs/harness/check_all.py`)

Action: Add a `.github/workflows/ci.yml` that runs on push and PR:

```

name: CI
on: [push, pull_request]
jobs:
  python-checks:
    runs-on: ubuntu-latest
    steps:
      - uses: actions/checkout@v4
      - uses: actions/setup-python@v5
        with:
          python-version: '3.12'
      - run: pip install ruff mypy python-can pywebview pyserial
      - run: ruff check bench/ apps/ rd-build/tools/ docs/build-research-hub.py
docs/harness/*.py
      - run: python -m py_compile bench/frames.py bench/can_bench.py apps/trackcluster-can-
sender/app.py
      - run: python -m pytest tests/ # once tests exist
  harness-checks:
    runs-on: ubuntu-latest
    steps:
      - uses: actions/checkout@v4
      - uses: actions/setup-python@v5
        with:
          python-version: '3.12'
      - run: python docs/harness/check_all.py

```

2.2 No dependency scanning

No Dependabot configuration exists. Dependencies are pinned with `>=` floors only, with no upper bounds or automated update PRs.

Action: Add `.github/dependabot.yml` :

```

version: 2
updates:
  - package-ecosystem: "pip"
    directory: "/bench"
    schedule:
      interval: "monthly"
  - package-ecosystem: "pip"
    directory: "/apps/trackcluster-can-sender"
    schedule:
      interval: "monthly"
  - package-ecosystem: "pip"
    directory: "/rd-build/tools"
    schedule:
      interval: "monthly"
  - package-ecosystem: "github-actions"
    directory: "/"
    schedule:
      interval: "monthly"

```

3. No Test Coverage (Priority: ● Medium)

3.1 Zero unit tests

None of the Python code has associated tests:

Module	Lines	Tests
bench/frames.py	351	0
bench/can_bench.py	806	0
apps/trackcluster-can-sender/app.py	333	0
docs/build-research-hub.py	1210	0
docs/harness/*.py (13 scripts)	~2000+	0

bench/frames.py is the highest-value test target — it contains the CAN frame encode/decode logic that must stay synchronized with `link_g4x_can_setup.json` and the frozen cluster firmware. A round-trip test (`encode` → `decode` → `verify`) for every frame would catch regressions immediately.

Action: Create `tests/test_frames.py` with round-trip tests for every `encode_*/decode_*` pair in `bench/frames.py`. Add boundary value tests (min/max raw values, edge cases like gear=7, empty payloads). This is the single most impactful improvement — it protects the core contract.

3.2 No contract validation test

There is no automated test that verifies `bench/frames.py` CAN IDs, scales, and offsets match `link_g4x_can_setup.json`. Currently, this is checked manually via the skills system and documentation rules.

Action: Create `tests/test_can_contract.py` that loads `link_g4x_can_setup.json` and asserts that every stream's ID, byte layout, scale, and offset match the corresponding constants and encode functions in `bench/frames.py`.

4. Documentation Drift (Priority: ● Medium)

4.1 CAN-CONFIG-STATUS.md violates handoff policy

File: `CAN-CONFIG-STATUS.md`

Issue: This is explicitly a "handoff note" (dated 2026-07-04), but `CLAUDE.md` states: "Handoffs are never committed. Work orders, handoffs and session summaries do not belong in this repo."

The file contains useful information (reconciliation results, corrected turbo speed scaling), but its format is wrong for a committed file.

Action: Extract the durable facts (boost=MAP not MGP, 2-byte oil/fuel pressure, turbo speed x1000 correction, cabin temp dropped) into the relevant permanent docs (`CAN-BUS-ID-ALLOCATION-TABLE.md` , `CAN-BUS-MASTER-DESIGN.md`). Then either delete `CAN-CONFIG-STATUS.md` or convert it to a permanent reference doc (not a "handoff note").

4.2 Stale references to removed apps

Files affected:

- `.gitignore` — comment references `apps/canbus-live-sender` (renamed to `trackcluster-can-sender`)
- `rd-build/PLAN.md` §6 — references `apps/canbus-bench-test.html` and `apps/canbus-live-sender` (both removed/renamed)
- `CLAUDE.md` "Known Gotchas" already notes the `AGENTS.md` reference to `apps/canbus-live-sender/` was fixed — verify this is still accurate

Action: Search-and-replace all references to `canbus-live-sender` → `trackcluster-can-sender` and `canbus-bench-test.html` → `bench/can_bench.py` across all documentation files.

4.3 AGENTS.md stale board reference

File: `AGENTS.md`

Issue: States "Claude progress board is a stale artifact (last updated 2026-09-01). Do not treat it as SoT." — this reference to a stale artifact should either be removed or the board should be deleted.

Action: Remove the stale board reference, or if the board file exists, delete it and remove the reference.

5. Dependency Management (Priority: ● Low)

5.1 No lock file / pinned versions

All three `requirements.txt` files use `>=` floors with no upper bounds:

File	Dependencies
<code>bench/requirements.txt</code>	<code>python-can>=4.0</code>
<code>apps/trackcluster-can-sender/requirements.txt</code>	<code>pywebview>=5.1</code> , <code>python-can[gs_usb]>=4.3.1</code> , <code>pyserial>=3.5</code> , <code>libusb-package>=1.0.26.2</code>
<code>rd-build/tools/requirements.txt</code>	<code>pyautogui>=0.9.54</code> , <code>pillow>=10.0.0</code>

A breaking change in any dependency (especially `python-can`, which has had breaking changes between 4.x minor versions) could silently break bench tests or the sender app.

Action: Add `pip-compile` (from `pip-tools`) or `uv lock` to generate pinned lock files (`requirements.lock` or `requirements.txt` with `==`). Add a `make lock` or `scripts/lock-deps.sh` command to regenerate.

5.2 Python version mismatch

CI uses Python 3.12 (`actions/setup-python@v5` with `python-version: '3.12'`), but local `__pycache__` contains `app.cpython-314.pyc` , indicating local development on Python 3.14. This mismatch could cause subtle behavior differences.

Action: Either bump CI to Python 3.13 (stable, widely available) or document the required Python version in a `pyproject.toml` `[project] requires-python = ">=3.12"` .

5.3 No pyproject.toml

The project has no `pyproject.toml` , meaning no centralized Python project configuration. Tool configuration (ruff, mypy, pytest) must be specified via CLI flags or separate config files.

Action: Add a minimal `pyproject.toml` at the repo root:

```
[tool.ruff]
line-length = 100
target-version = "py312"

[tool.ruff.lint]
select = ["E", "F", "W", "I", "UP"]

[tool.mypy]
python_version = "3.12"
ignore_missing_imports = true

[tool.pytest.ini_options]
testpaths = ["tests"]
```

6. Code Quality (Priority: Low)

6.1 Type hint inconsistencies

bench/frames.py:

- `decode_*` functions return `Dict[str, float]` but some values are `int` (gear, fuel_level_pct, heartbeat, etc.)
- Should be `Dict[str, float | int]` or separate return types

bench/can_bench.py:

- `Phase.apply` uses `Callable[[object, float], None]` — should use a `Protocol` or proper type parameter
- `ClusterState` and `RealDashState` are plain classes but would benefit from `@dataclass`
- `cmd_monitor` uses bare `dict` without type parameters (`counts: dict = {}`)

apps/trackcluster-can-sender/app.py:

- `Api` class has bare `except Exception` in 8+ places — could mask real bugs
- `_adapters` is read outside the lock in `get_state()` — potential race condition (minor, since it's just for UI display)

6.2 No formatter configured

No `black`, `ruff format`, or equivalent is configured. Code style is consistent (suggesting manual discipline), but automated formatting would prevent drift.

Action: Add `ruff format` to the CI workflow and a pre-commit hook.

6.3 docs/build-research-hub.py is a 1210-line monolith

This single script handles HTML parsing, markdown parsing, CSV reading, table extraction, git date lookup, and HTML generation. It would benefit from being split into modules (parser, renderer, data model).

Action: Low priority — only refactor if the script needs new features. Currently works well and is well-commented.

7. Security & Safety (Priority: Minimal)

7.1 PyAutoGUI FAILSAFE disabled

File: rd-build/tools/automation_helper.py

Line: pyautogui.FAILSAFE = False

This disables PyAutoGUI's fail-safe (mouse-to-corner-to-abort). For an automation script driving a GUI editor, this is a known risk — if the script goes wrong, there's no easy way to stop it short of killing the process.

Action: Document this as intentional (the script is agent-driven and may move the mouse to screen corners during normal operation). Add a comment explaining the rationale. Consider adding a `Ctrl-C` handler that restores the fail-safe.

7.2 No secrets exposure

The `.gitignore` properly excludes `rd-build/CREDENTIALS.md` and `rd-build/shots/`. No API keys, tokens, or credentials were found in committed code.

Prioritized Action Plan

Phase 1: Quick Wins (1-2 hours, high impact)

#	Action	Effort	Impact
1	<code>git rm -r --cached</code> all <code>__pycache__</code> directories	5 min	Removes tracked build artifacts
2	Delete 9 versioned <code>st185_dash_v*.rd</code> files from <code>rd-build/realdash-root/</code>	5 min	Removes 9 stale binary files
3	Fix stale <code>canbus-live-sender</code> references in <code>.gitignore</code> and <code>rd-build/PLAN.md</code>	15 min	Eliminates confusing dead references
4	Add <code>pyproject.toml</code> with <code>ruff</code> + <code>mypy</code> config	15 min	Enables future tooling
5	Add <code>.github/dependabot.yml</code>	10 min	Automated dependency updates

Phase 2: CI & Tests (4-8 hours, high impact)

#	Action	Effort	Impact
6	Add <code>.github/workflows/ci.yml</code> with <code>ruff</code> + <code>py_compile</code>	1 hr	Catches lint/compile errors on every PR
7	Create <code>tests/test_frames.py</code> with round-trip encode/decode tests	2-3 hr	Protects the core CAN contract
8	Create <code>tests/test_can_contract.py</code> validating <code>frames.py</code> ↔ JSON parity	1-2 hr	Catches cross-file drift automatically
9	Add harness <code>check_all.py</code> to CI	30 min	Validates wiring on every PR

Phase 3: Documentation Cleanup (2-4 hours, medium impact)

#	Action	Effort	Impact
10	Convert CAN-CONFIG-STATUS.md from handoff to permanent reference, or delete and merge facts into canonical docs	1 hr	Fixes policy violation
11	De-duplicate CLAUDE.md and AGENTS.md shared sections	1 hr	Reduces maintenance burden
12	Review and archive/delete rd-build/tools/ one-off scripts	30 min	Removes stale tooling
13	Remove stale board reference from AGENTS.md	5 min	Removes confusing stale info

Phase 4: Modernization (4-8 hours, low urgency)

#	Action	Effort	Impact
14	Add ruff format to CI and format all Python files	1 hr	Consistent style enforcement
15	Fix type hint inconsistencies in frames.py and can_bench.py	1 hr	Better IDE support and catch type errors
16	Generate pinned dependency lock files	30 min	Reproducible installs
17	Bump CI Python version or document required version	15 min	Eliminates version mismatch
18	Add bare except Exception logging to sender app	30 min	Better error diagnostics

Dependency Summary

Dependency	Current Floor	Latest Stable	Status
python-can	>=4.0	4.5.x	✅ Fine, consider bumping to >=4.3
pywebview	>=5.1	5.x	✅ Current
pyserial	>=3.5	3.5	✅ Current (no 4.x yet)
libusb-package	>=1.0.26.2	1.0.26.x	✅ Current
pyautogui	>=0.9.54	0.9.54	✅ Current
pillow	>=10.0.0	11.x	⚠️ Could bump to >=11.0
actions/checkout	v4	v4	✅ Current
actions/setup-python	v5	v5	✅ Current
actions/upload-artifact	v4	v4	✅ Current

Files Examined

File	Purpose
bench/frames.py	CAN frame encode/decode (351 lines)
bench/can_bench.py	CAN bench test tool (806 lines)
bench/requirements.txt	Bench deps (1 dep)
apps/trackcluster-can-sender/app.py	Desktop CAN sender app (333 lines)
apps/trackcluster-can-sender/requirements.txt	Sender app deps (4 deps)
docs/build-research-hub.py	Research hub HTML generator (1210 lines)
docs/harness/check_all.py	Harness validation gate (52 lines)
docs/harness/*.py	13 harness audit scripts
rd-build/tools/automation_helper.py	RealDash editor automation (151 lines)
rd-build/tools/requirements.txt	Automation deps (2 deps)
link_g4x_can_setup.json	Canonical CAN config (130 lines)
.github/workflows/research-hub.yml	Only CI workflow (84 lines)
CLAUDE.md	Agent instructions (143 lines)
AGENTS.md	Agent instructions (149 lines)
CAN-CONFIG-STATUS.md	Stale handoff note (50 lines)
.gitignore, .gitattributes	Repo config
tune/engine_constants.yaml	Engine calibration data
GitHub remote	0 open issues, 20 recent commits reviewed

Generated 2026-09-27. This is a point-in-time analysis based on the repo state at commit `b265f1d` .